

CSE 432 Robotics

Rigid Motions and Homogeneous Transformations 2

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PARAMETERIZATIONS OF ROTATIONS

- A rigid body possesses at most three rotational 3 DOF and thus at most three quantities are required to specify its orientation.
- The nine elements r_{ij} in a general rotational transformation R are not independent quantities.
- The elements r_{ij} of the rotation matrices should satisfy the following constraints:

$$\sum_i r_{ij}^2 = 1, \quad j \in \{1,2,3\}$$
$$r_{1i}r_{1j} + r_{2i}r_{2j} + r_{3i}r_{3j} = 0, \quad i \neq j$$



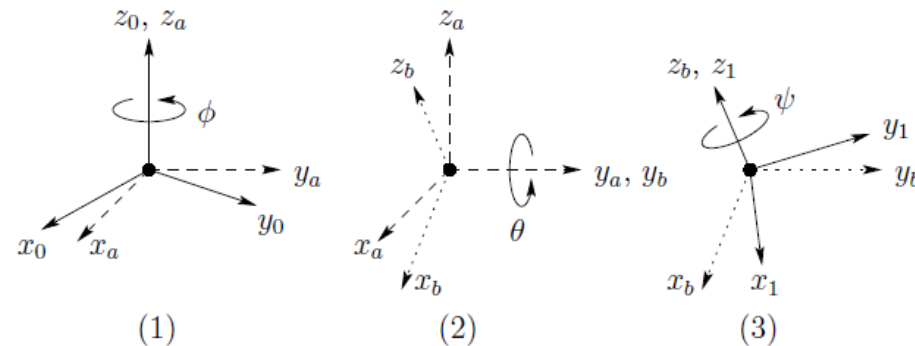
Euler Angles

- A common way to specify a rotation matrix using three parameters is through Euler angles.
- There exist twelve possible sequences of rotation axes
$$\begin{array}{lll} z - x - z, & x - y - x, & y - z - y, \\ z - y - z, & x - z - x, & y - x - y, \\ x - y - z, & y - z - x, & z - x - y, \\ x - z - y, & z - y - x, & y - x - z \end{array}$$
- In this course, the sequence $z - y - z$ is the default sequence.
- All rotation are **about the current frame**.



Euler Angles

- Consider the fixed coordinate frame $o_0x_0y_0z_0$ and the rotated frame $o_1x_1y_1z_1$.
- We can specify the orientation of the frame $o_1x_1y_1z_1$ relative to the frame $o_0x_0y_0z_0$ by three angles (ϕ, θ, ψ) , known as Euler Angles, and obtained by three successive rotations as follows:
 - First rotate about the z-axis by the angle ϕ .
 - Next rotate about the current y-axis by the angle θ .
 - Finally rotate about the current z-axis by the angle ψ .





Euler Angles

- In terms of the basic rotation matrices the resulting rotational transformation can be generated as the product

$$\begin{aligned} R_{ZYZ} &= R_{z,\phi} R_{y,\theta} R_{z,\psi} = \begin{bmatrix} c\phi & -s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix} \begin{bmatrix} c\psi & -s\psi & 0 \\ s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c\phi c\theta c\psi - s\phi s\psi & -c\phi c\theta s\psi - s\phi c\psi & c\phi s\theta \\ s\phi c\theta c\psi + c\phi s\psi & -s\phi c\theta s\psi + c\phi c\psi & s\phi s\theta \\ -s\theta c\psi & s\theta s\psi & c\theta \end{bmatrix} \\ &= \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \end{aligned}$$



Euler Angles

$$R_{ZYZ} = \begin{bmatrix} c\phi c\theta c\psi - s\phi s\psi & -c\phi c\theta s\psi - s\phi c\psi & c\phi s\theta \\ s\phi c\theta c\psi + c\phi s\psi & -s\phi c\theta s\psi + c\phi c\psi & s\phi s\theta \\ -s\theta c\psi & s\theta s\psi & c\theta \end{bmatrix}$$

- First, suppose that not both of r_{13}, r_{23} are zero $\rightarrow s_\theta \neq 0$, then $r_{33} \neq \pm 1$, and we have

$$c_\theta = r_{33}, \quad s_\theta = \pm \sqrt{1 - r_{33}^2}$$

- Thus,

$$\theta = \text{atan2} \left(r_{33}, \sqrt{1 - r_{33}^2} \right) \quad (I), \text{ OR}$$

$$\theta = \text{atan2} \left(r_{33}, -\sqrt{1 - r_{33}^2} \right) \quad (II)$$



Euler Angles

$$R_{ZYZ} = \begin{bmatrix} c\phi c\theta c\psi - s\phi s\psi & -c\phi c\theta s\psi - s\phi c\psi & c\phi s\theta \\ s\phi c\theta c\psi + c\phi s\psi & -s\phi c\theta s\psi + c\phi c\psi & s\phi s\theta \\ -s\theta c\psi & s\theta s\psi & c\theta \end{bmatrix}$$

- If we choose the value for θ given by Equation I, then $s_\theta > 0$, and
$$\phi = \text{atan2}(r_{13}, r_{23})$$
$$\psi = \text{atan2}(-r_{31}, r_{32})$$
- If we choose the value for θ given by Equation II, then $s_\theta < 0$, and
$$\phi = \text{atan2}(-r_{13}, -r_{23})$$
$$\psi = \text{atan2}(r_{31}, -r_{32})$$
- **There are two solutions depending on the sign chosen for θ .**



Euler Angles

- If $r_{13} = r_{23} = 0$, then the fact that R is orthogonal implies that $r_{33} = \pm 1$, and that $r_{31} = r_{32} = 0$. Thus, R has the form

$$R_{ZYZ} = \begin{bmatrix} c\phi c\theta c\psi - s\phi s\psi & -c\phi c\theta s\psi - s\phi c\psi & c\phi s\theta \\ s\phi c\theta c\psi + c\phi s\psi & -s\phi c\theta s\psi + c\phi c\psi & s\phi s\theta \\ -s\theta c\psi & s\theta s\psi & c\theta \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & 0 \\ r_{21} & r_{22} & 0 \\ 0 & 0 & \pm 1 \end{bmatrix}$$

- If $r_{33} = 1$, then $c\theta = 1$ and $s\theta = 0$, so that $\theta = 0$.
- Thus, the sum $\phi + \psi$ can be determined as
$$\phi + \psi = \text{atan2}(r_{11}, r_{21}) = \text{atan2}(r_{11}, -r_{12})$$
- If $r_{33} = -1$, then $c = -1$ and $s = 0$, so that $\theta = \pi$.
- Thus, the $\phi - \psi$ can be determined as
$$\phi - \psi = \text{atan2}(-r_{11}, -r_{12})$$
- **Hence, there are infinite solutions.**



Example

- Find R corresponding to the set of Euler angles $\{0, \frac{\pi}{2}, \frac{\pi}{4}\}$.
- What is the direction of the x_1 axis relative to the base frame?

$$R_1^0 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- The direction of the x_1 axis relative to the base frame

$$x_1 = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$



Atan2 Function

- $\text{Atan2}(x,y)$ returns the arc tangent of the two numbers x and y .
- It calculate the arc tangent of y/x , except that the signs of both arguments are used to determine the quadrant of the result.

$$\text{atan2}(x, y) = \begin{cases} \arctan\left(\frac{y}{x}\right), & \text{if } x > 0 \\ \arctan\left(\frac{y}{x}\right) + \pi, & \text{if } x < 0 \text{ and } y \geq 0 \\ \arctan\left(\frac{y}{x}\right) - \pi, & \text{if } x < 0 \text{ and } y < 0 \\ +\frac{\pi}{2}, & \text{if } x = 0 \text{ and } y > 0 \\ -\frac{\pi}{2}, & \text{if } x = 0 \text{ and } y < 0 \\ \text{undefined}, & \text{if } x = 0 \text{ and } y = 0 \end{cases}$$



Example

- Find the rotation matrix corresponding to the set of Euler angles $\left\{\frac{\pi}{2}, 0, \frac{\pi}{4}\right\}$.
- What is the direction of the x_1 axis relative to the base frame?

$$R_1^0 = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- The direction of the x_1 axis relative to the base frame

$$x_1 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$$



Roll, Pitch, Yaw Angles

- A rotation matrix R can also be described as a product of successive rotations about the principal coordinate axes x_0, y_0 , and z_0 “**Fixed Frame**” taken in a specific order.
- These rotations define the **roll**, **pitch**, and **yaw** angles, which also denoted ϕ, θ, ψ .
- **Rotation Order:** $x \rightarrow y \rightarrow z$
 - Yaw about x_0 by angle ψ
 - Pitch about y_0 by angle θ
 - Roll about z_0 by angle ϕ

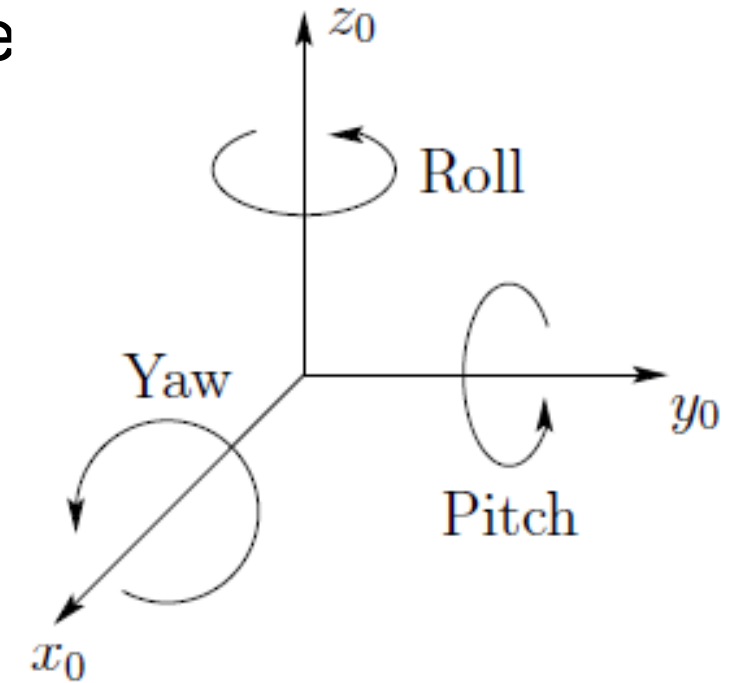


Roll, Pitch, Yaw Angles

- Rotations are relative to the **fixed frame**, the resulting transformation matrix:

$$R = R_{z,\phi} R_{y,\theta} R_{x,\psi}$$

$$\begin{aligned} &= \begin{bmatrix} c_\phi & -s_\phi & 0 \\ s_\phi & c_\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\psi & -s_\psi \\ 0 & s_\psi & c_\psi \end{bmatrix} \\ &= \begin{bmatrix} c_\phi c_\theta & -s_\phi c_\psi + c_\phi s_\theta s_\psi & s_\phi s_\psi + c_\phi s_\theta c_\psi \\ s_\phi c_\theta & c_\phi c_\psi + s_\phi s_\theta s_\psi & -c_\phi s_\psi + s_\phi s_\theta c_\psi \\ -s_\theta & c_\theta s_\psi & c_\theta c_\psi \end{bmatrix} \end{aligned}$$





Axis/Angle Representation

- Rotations are not always performed about the principal coordinate axes.
- We are often interested in a rotation about an arbitrary axis in space.
- This provides both a convenient way to describe rotations, and an alternative parameterization for rotation matrices.
- Let $k = (k_x, k_y, k_z)$, expressed in the frame $o_0x_0y_0z_0$, be a unit vector defining an axis.
- We wish to derive the rotation matrix $R_{k,\theta}$, representing a rotation of θ about this axis.



Axis/Angle Representation

- One approach: align the axis k with the **z-axis** using rotations.

- Use the transformation:

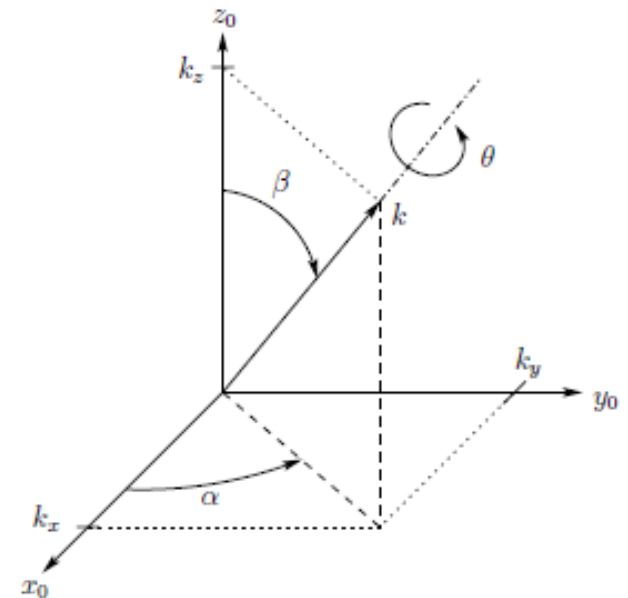
$$R_1^0 = R_{z,\alpha} R_{y,\beta}$$

- Compute the rotation about k using a **similarity transform**:

$$R_{k,\theta} = R_1^0 R_{z,\theta} (R_1^0)^{-1}$$

- **Expanded form:**

$$R_{k,\theta} = R_{z,\alpha} R_{y,\beta} R_{z,\theta} R_{y,-\beta} R_{z,-\alpha}$$





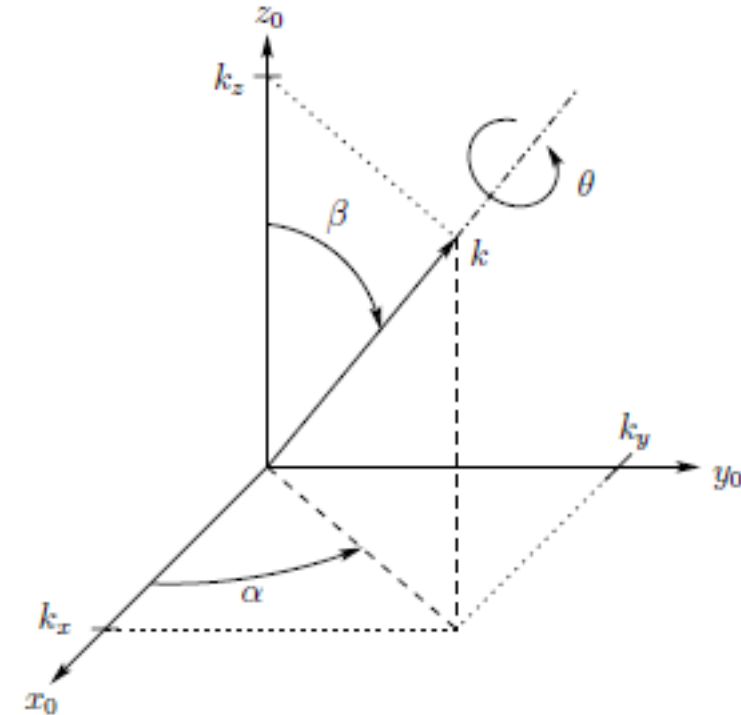
Axis/Angle Representation

- We can obtain

$$\sin \alpha = \frac{k_y}{\sqrt{k_x^2 + k_y^2}}$$

$$\cos \alpha = \frac{k_x}{\sqrt{k_x^2 + k_y^2}}$$

$$\sin \beta = \sqrt{k_x^2 + k_y^2}$$
$$\cos \beta = k_z$$



- **The last two equations follow from k being a unit vector.**



Axis/Angle Representation

- We obtain after some lengthy calculation

$$R_{k,\theta} = \begin{bmatrix} k_x^2 v_\theta + c_\theta & k_x k_y v_\theta - k_z s_\theta & k_x k_z v_\theta + k_y s_\theta \\ k_x k_y v_\theta + k_z s_\theta & k_y^2 v_\theta + c_\theta & k_y k_z v_\theta - k_x s_\theta \\ k_x k_z v_\theta - k_y s_\theta & k_y k_z v_\theta + k_x s_\theta & k_z^2 v_\theta + c_\theta \end{bmatrix}$$

- where $v_\theta = 1 - \cos \theta$
- Any rotation matrix can be represented as a single rotation about some axis by an appropriate angle,

$$R = R_{k,\theta}$$

- where k is a unit vector defining the axis of rotation, and θ is the angle of rotation about k .
- The matrix $R_{k,\theta}$, is called the axis/angle representation of R .



Axis/Angle Representation

- Given an arbitrary rotation matrix R with components r_{ij} , the equivalent angle θ and equivalent axis k are given by the expressions

$$\theta = \cos^{-1} \left(\frac{\text{Tr}(R) - 1}{2} \right) = \cos^{-1} \left(\frac{r_{11} + r_{22} + r_{33} - 1}{2} \right)$$

- where Tr denotes the trace of R , and

$$k = \frac{1}{2 \sin \theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$



Example 9

- Suppose R is generated by a rotation of 90° about z_0 followed by a rotation of 30° about y_0 followed by a rotation of 60° about x_0 . Then

$$R = R_{x,60}R_{y,30}R_{z,90} = \begin{bmatrix} 0 & -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{4} & -\frac{3}{4} \\ \frac{\sqrt{3}}{2} & \frac{1}{4} & \frac{\sqrt{3}}{4} \end{bmatrix}$$

- $Tr(R) = 0 \rightarrow$ the equivalent angle and axis are given as

$$\theta = \cos^{-1}\left(-\frac{1}{2}\right) = 120^\circ, \quad k = \left(\frac{1}{\sqrt{3}}, \frac{1}{2\sqrt{3}}, -\frac{1}{2}, \frac{1}{2\sqrt{3}} + \frac{1}{2}\right)$$



RIGID MOTIONS

- Rigid motion is a pure translation together with a pure rotation.
- Let R_1^0 be the rotation matrix that specify the orientation of frame $o_1x_1y_1z_1$ with respect to frame $o_0x_0y_0z_0$ and d be the vector from the origin of frame $o_0x_0y_0z_0$ to the origin of frame $o_1x_1y_1z_1$.
- Then the coordinates p^0 are given by
$$p^0 = R_1^0 p^1 + d^0$$



RIGID MOTIONS

- The rigid motions can be represented by the set of matrices of the form

$$H = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix}; R \in SO(3), d \in \mathbb{R}^3$$

- Using the fact that R is orthogonal it is an easy exercise to show that the inverse transformation H^{-1} is given by

$$H^{-1} = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix}$$



HOMOGENEOUS TRANSFORMATIONS

- To represent the transformation by a matrix multiplication, we must augment the vectors by the addition of a fourth component of 1 as follows

$$P^0 = \begin{bmatrix} p^0 \\ 1 \end{bmatrix}$$

- The vectors P^0 are known as homogeneous representations of the vectors p^0

$$P^0 = H_1^0 P^1$$



HOMOGENEOUS TRANSFORMATIONS

- A set of basic homogeneous transformations is given by

$$Trans_{x,a} = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, Rot_{x,\alpha} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Trans_{y,b} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, Rot_{y,\beta} = \begin{bmatrix} \cos \beta & 0 & \sin \beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \beta & 0 & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Trans_{z,c} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix}, Rot_{z,\gamma} = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 & 0 \\ \sin \gamma & \cos \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Composition Rule for Homogeneous Transformations

- The same interpretation regarding composition and ordering of transformations holds for 4×4 homogeneous transformations as for 3×3 rotations.
- Given a homogeneous transformation H_1^0 relating two frames, if a second rigid motion, represented by H is performed relative to the current frame, then

$$H_2^0 = H_1^0 H$$

- whereas if the second rigid motion is performed relative to the fixed frame, then

$$H_2^0 = H H_1^0$$



Example 10

- The homogeneous transformation matrix H that represents a
 - rotation by angle α about the current x-axis followed by
 - a translation of b units along the current x-axis, followed by
 - a translation of d units along the current z-axis, followed by
 - a rotation by angle θ about the current z-axis,

- The homogeneous transformation is given by

$$H = Rot_{x,\alpha} Trans_{x,b} Trans_{z,d} Rot_{z,\theta}$$

$$H = \begin{bmatrix} c\theta & -s\theta & 0 & b \\ c\alpha s\theta & c\alpha c\theta & -s\alpha & -ds\alpha \\ s\alpha s\theta & s\alpha c\theta & c\alpha & dc\alpha \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Any Questions